



B. Tech. in INFORMATION TECHNOLOGY
III Year II Semester Syllabus (KR23)
INTERNET OF THINGS (23IT602PC)
Open Elective-I

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Pre-requisite:

1. 23CS203ES-Python Programming

Course Objectives: The objectives of the course are to:

1. Understand the concepts of Internet of Things and able to build IoT applications
2. Understand the need of Sensors & Actuators in IoT
3. Learn the programming and use of Arduino boards.
4. Learn Raspberry Pi python programming and use of Raspberry Pi boards.
5. Learn interfacing I/O devices with of Raspberry Pi boards.

Course Outcomes: After learning the contents of this course the student is be able to

1. Describe basic concepts of Internet of Things in real world applications.
2. Identify the need and working of Sensors & Actuators in IoT
3. Practice Programs and configure Arduino boards for various designs.
4. Practice Raspberry Pi Python programming and interfacing I/O devices with Raspberry Pi.
5. Design IoT applications in different domains using Raspberry Pi.

UNIT-I

Introduction to IOT: Definition & Characteristics of IoT, Physical design of IoT, Logical design of IoT, Functional blocks of IoT, Communication models & APIs, IoT Enabling Technologies

UNIT -II

IoT Sensing & Actuation: Introduction to sensors, sensor characteristics, sensorial deviations, sensing types, Temperature & Humidity Sensor, PIR Sensor, Ultrasonic Sensor, LDR, sensing considerations, Actuators, Actuator types, LED, Switch, Pi camera, Actuator considerations

UNIT -III

Arduino Programming: This is Arduino, Getting strated ,Arduino Programming: variables, Conditional execution, loops, functions, arrays and strings, , input and output, the serial monitor, Integration of Sensors and Actuators with Arduino: Controlling LED ,integration of an LED and Switch, integration of an LED and light sensor (LDR), Interfacing LCD .

UNIT IV

IoT Physical Devices & Endpoints: what is an IoT Device: Basic building blocks of an IoT device, Introduction to Raspberry: Raspberry Pi Board, Raspberry Pi Interfaces (serial, SPI, I2C), Programming Raspberry Pi with Python: Controlling LED, integration of LED and Switch, interfacing an Ultrasonic Sensor, PIR sensor, DHT22 sensor, Pi camera.

UNIT- V

Implementation of IoT with Raspberry Pi:

Implementation of data sharing between server & client, Traffic light control system, Domain Specific Applications of IoT applications: implementation of Home automation, Healthcare, Smart device, Agriculture

TEXT BOOKS:

7. Internet of Things - A Hands-on Approach, Arshdeep Bahga and Vijay Madisetti, Universities Press, 2015.
8. "Introduction to IoT", Sudip Misra, Anandarup Mukherjee, Arijit Roy, Cambridge University Press, 2021
9. "Introduction to Internet of Things", Sudip Misra, IIT Kharagpur
https://onlinecourses.nptel.ac.in/noc22_cs53/unit?unit=113&lesson=114
10. Programming Arduino Next Steps: Going Further with Sketches, Simon Monk, Second Edition, 2019.
11. Getting Started with Raspberry Pi, Matt Richardson & Shawn Wallace, O'Reilly (SPD), 2014.
12. <https://projects.raspberrypi.org>

Reference Books:

4. Beginning Sensor networks with Arduino and Raspberry Pi – Charles Bell, Apress, 2013.
5. "The Internet of Things: Enabling Technologies, Platforms, and Use Cases", by Pethuru Raj and Anupama C. Raman (CRC Press)
6. <https://www.arduino.cc>